TOPSTECHNOLOGIES
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Comptia N+ Assignment

[N+ - Network security, Maintenance and Troubleshooting procedures]

Software and Hardware Tools

Task:

❖ **1. Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.**

➢ **Aim**

- To install Wireshark on a laptop, capture two minutes of network traffic while browsing Instagram or YouTube, save the captured traffic in .pcap format, and identify the top three protocols.

➢ **Introduction**

- **Wireshark** is a free and open-source network protocol analyzer used to capture and examine network traffic.
- It allows users to observe packets travelling through a network and identify the protocols involved in communication.
- When we browse websites such as Instagram or YouTube, our computer communicates with different servers using protocols such as DNS, TCP, TLS, UDP, and QUIC.

- In this practical, Wireshark is used to capture network traffic for two minutes and identify the top three protocols.

➢ **Hardware and Software Requirements**

➢ **Hardware Requirements**

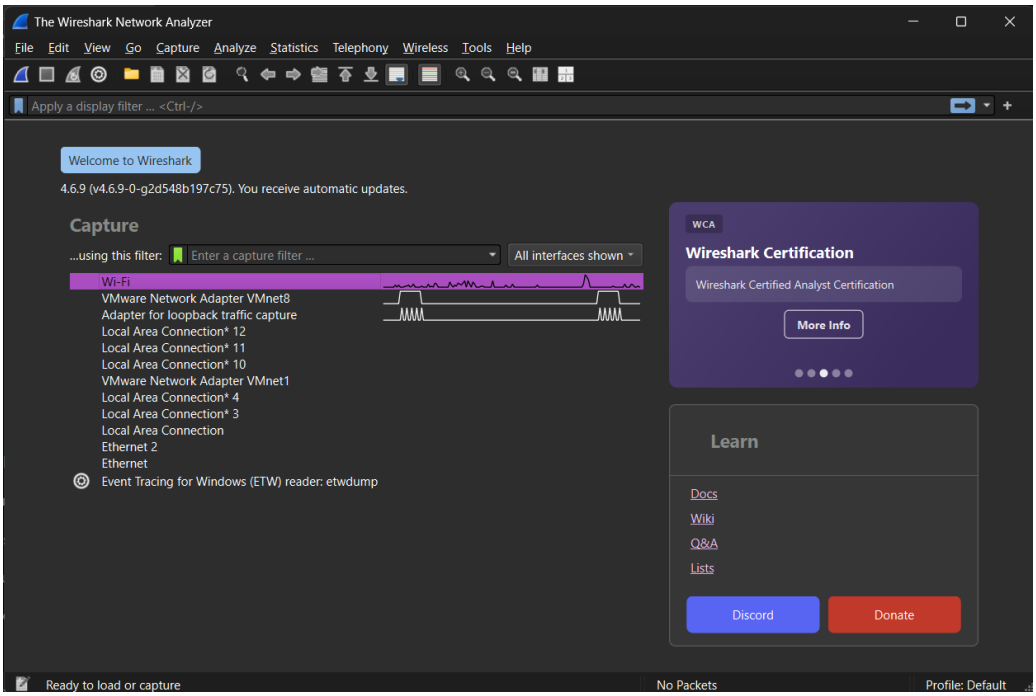
- Laptop or desktop computer
- Wi-Fi or Ethernet connection
- Sufficient storage space

➢ **Software Requirements**

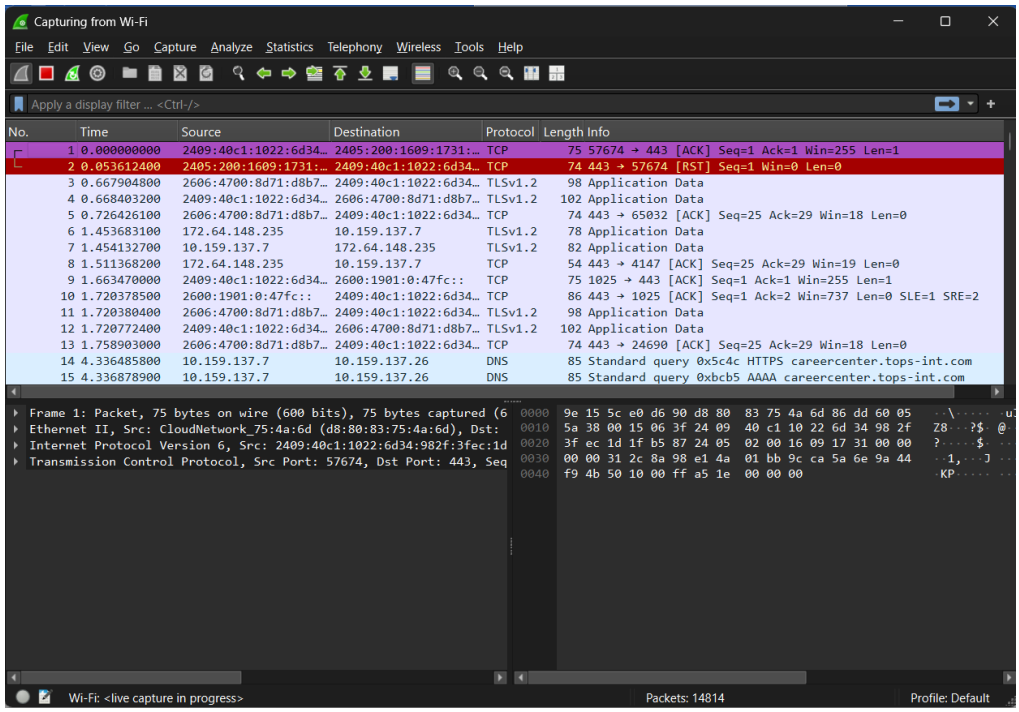
- Wireshark software
- Windows operating system
- Web browser such as Google Chrome or Microsoft Edge
- Internet connection

➢ **Procedure**

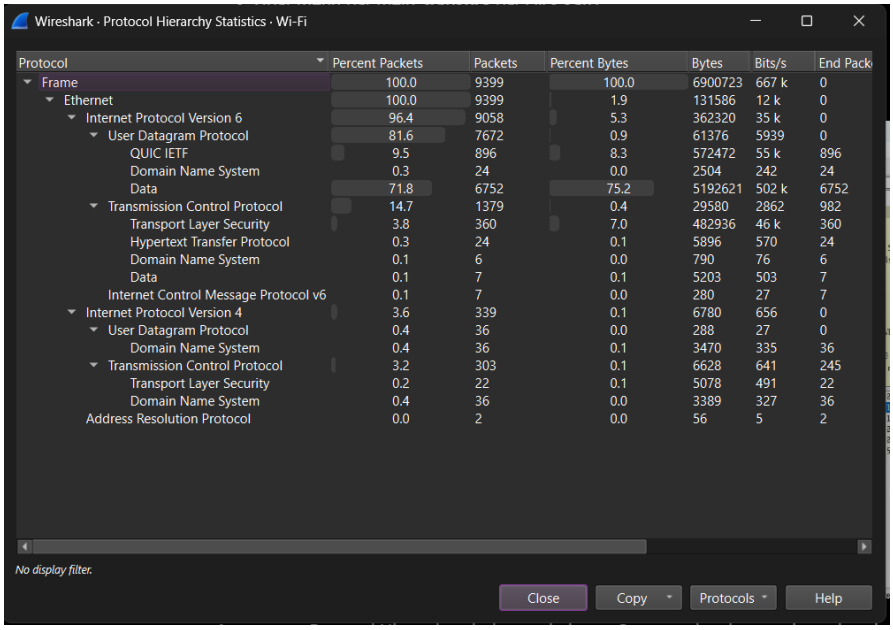
1. Download and install Wireshark from its official website.
2. Restart the computer if required to complete the installation.
3. Open Wireshark and select the active network interface, such as Wi-Fi or Ethernet.



4. Start capturing network packets.

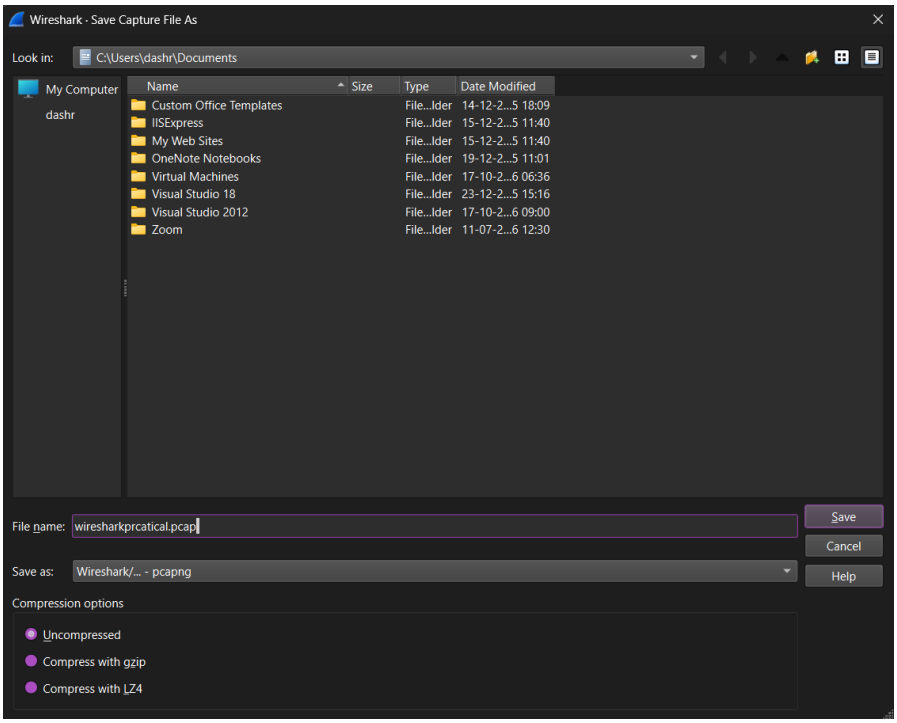


5. Open a web browser and visit www.instagram.com or www.youtube.com.
6. Browse the website for approximately two minutes.
7. Return to Wireshark and stop the packet capture.
8. Open **Statistics → Protocol Hierarchy** to examine the protocols and their packet counts.



Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Pack
Frame	100.0	9399	100.0	6900723	667 k	0
Ethernet	100.0	9399	1.9	131586	12 k	0
Internet Protocol Version 6	96.4	9058	5.3	362320	35 k	0
User Datagram Protocol	81.6	7672	0.9	61376	5939	0
QUIC IETF	9.5	896	8.3	572472	55 k	896
Domain Name System	0.3	24	0.0	2504	242	24
Data	71.8	6752	75.2	5192621	502 k	6752
Transmission Control Protocol	14.7	1379	0.4	29580	2862	982
Transport Layer Security	3.8	360	7.0	482936	46 k	360
Hypertext Transfer Protocol	0.3	24	0.1	5896	570	24
Domain Name System	0.1	6	0.0	790	76	6
Data	0.1	7	0.1	5203	503	7
Internet Control Message Protocol v6	0.1	7	0.0	280	27	7
Internet Protocol Version 4	3.6	339	0.1	6780	656	0
User Datagram Protocol	0.4	36	0.0	288	27	0
Domain Name System	0.4	36	0.1	3470	335	36
Transmission Control Protocol	3.2	303	0.1	6628	641	245
Transport Layer Security	0.2	22	0.1	5078	491	22
Domain Name System	0.4	36	0.0	3389	327	36
Address Resolution Protocol	0.0	2	0.0	56	5	2

9. Identify the three protocols with the highest packet counts.
10. Select **File → Save As** and save the capture in .pcap format.



- **Observation**
- During the two-minute capture, Wireshark displays different network packets generated while browsing the selected website.
- **Top Three Protocols Observed**

Sr. No.	Protocol Name	Packet Count
1	IPv6 (Internet Protocol Version 6)	9058
2	UDP (User Datagram Protocol)	7672

3	TCP (Transmission Control Protocol)	1379

- **Result**
- Wireshark was successfully installed and used to capture approximately two minutes of network traffic while browsing Instagram or YouTube. The captured traffic was saved in .pcap format, and the top three protocols were identified.

❖ **2. Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.**

- **Aim**
  - To use the ping command to test network connectivity to www.spotify.com and www.zomato.com, record their response times, and identify which website responds faster.

- **Introduction**
  - **Ping (Packet Internet Groper)** is a network troubleshooting command used to check whether a destination device or server is reachable over a network.
  - It sends ICMP Echo Request packets to the destination and waits for ICMP Echo Reply packets.

- The response time is measured in milliseconds (ms). A lower average response time generally indicates a faster network response.
- In this practical, the ping command is used to test connectivity to Spotify and Zomato and compare their response times.

➢ **Requirements**

- A computer or laptop
- Internet connection
- Command Prompt (Windows) or Terminal (Linux/Mac)

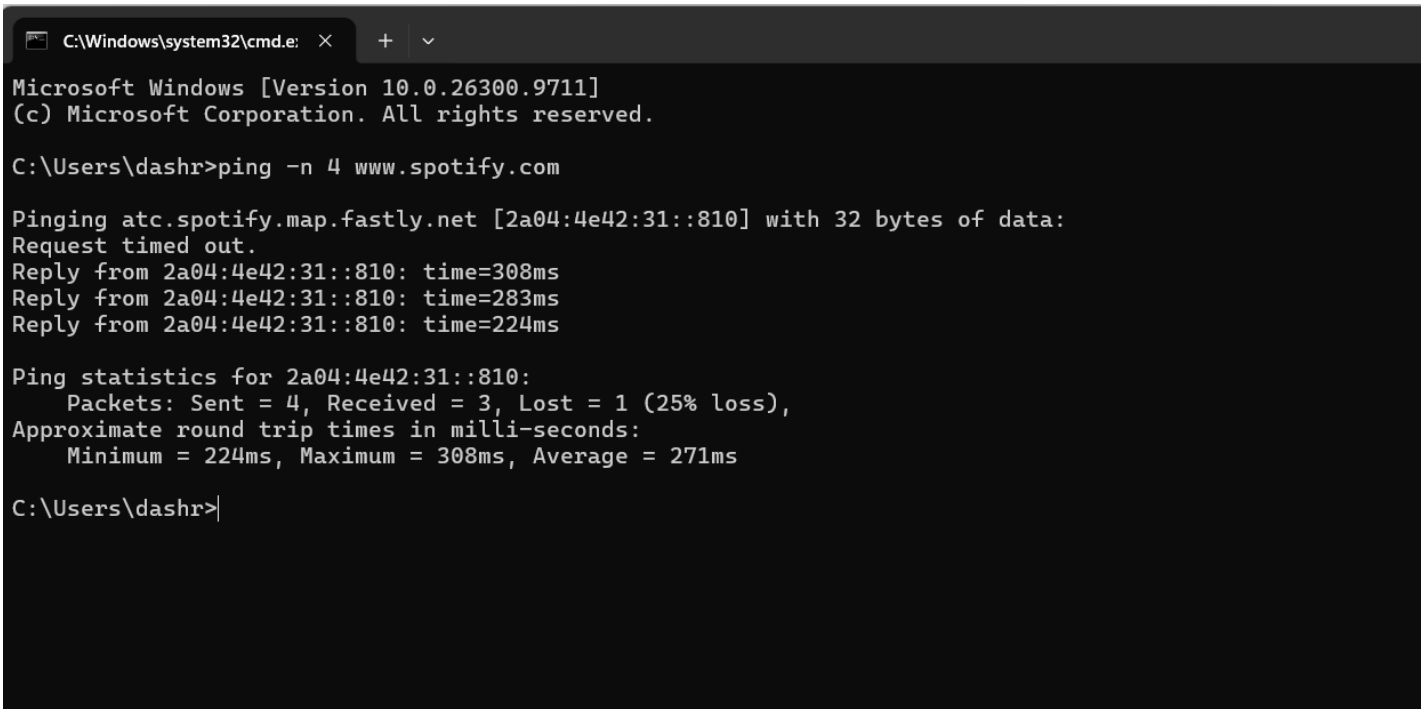
➢ **Procedure**

➢ **Step 1: Open Command Prompt**

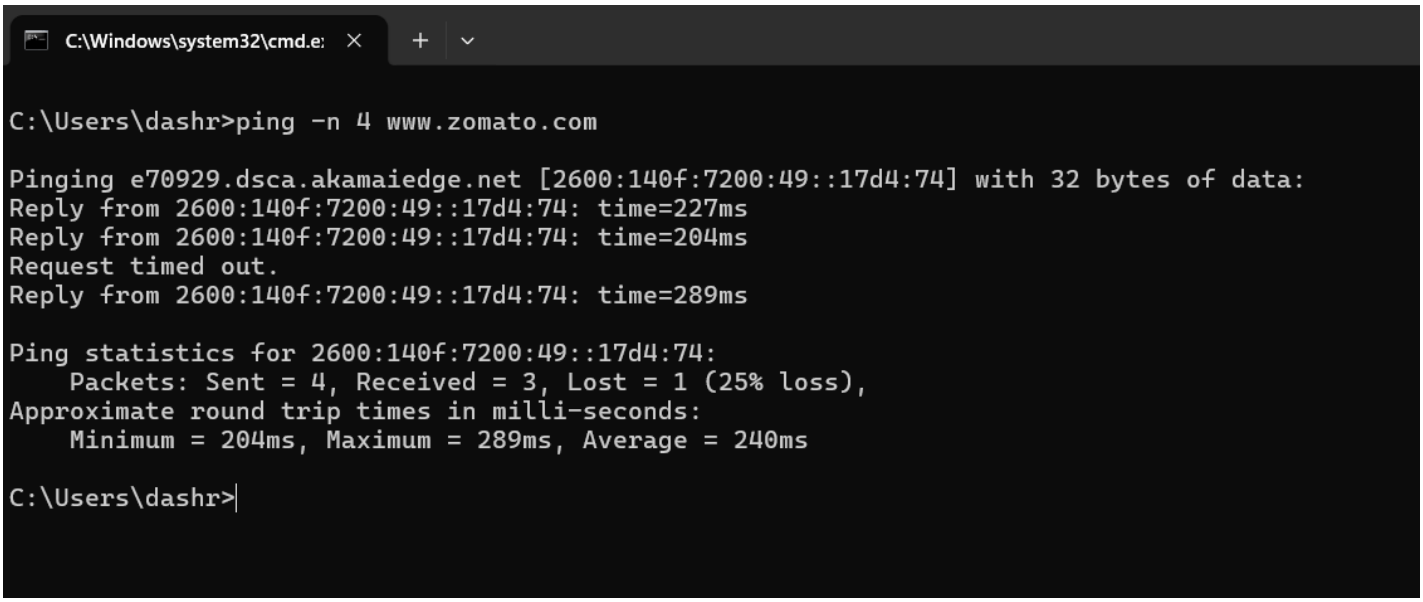
1. Press Windows + R.
2. Type cmd.
3. Press Enter to open Command Prompt.

➢ **Step 2: Test Connectivity to Spotify**

1. Enter the following command:
   ping -n 4 www.spotify.com
2. Press Enter.
3. Wait for the command to complete.
4. Observe the reply messages and record the average response time.



- **Step 3: Test Connectivity to Zomato**
  1. Enter the following command:
     ping -n 4 www.zomato.com
  2. Press Enter.
  3. Wait for the command to complete.
  4. Observe the reply messages and record the average response time.



- **Step 4: Compare Response Times**
- Compare the average response times of both websites. The website with the lower average response time is considered to have responded faster in this test.

- **Observation**
- Record the actual results obtained from the Command Prompt.

Particulars	Spotify	Zomato
Website	www.spotify.com	www.zomato.com
Command	ping -n 4 www.spotify.com	ping -n 4 www.zomato.com
Packets Sent	4	4
Packets Received	3	3

Packet Loss	1	1
Average Response Time	271 ms	240 ms

- **Comparison**
  - Website with lower average response time: Zomato (240 ms)
  - Website that responded faster: Zomato
- **Result**
- The ping command was used to test connectivity to www.spotify.com and www.zomato.com. Their average response times were recorded and compared to identify which website responded faster.

❖ **3. Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.**

➢ **Aim**

➢ To use Windows Task Manager to identify running processes related to Google Chrome, WhatsApp Desktop, or other browser-based applications and record their memory usage.

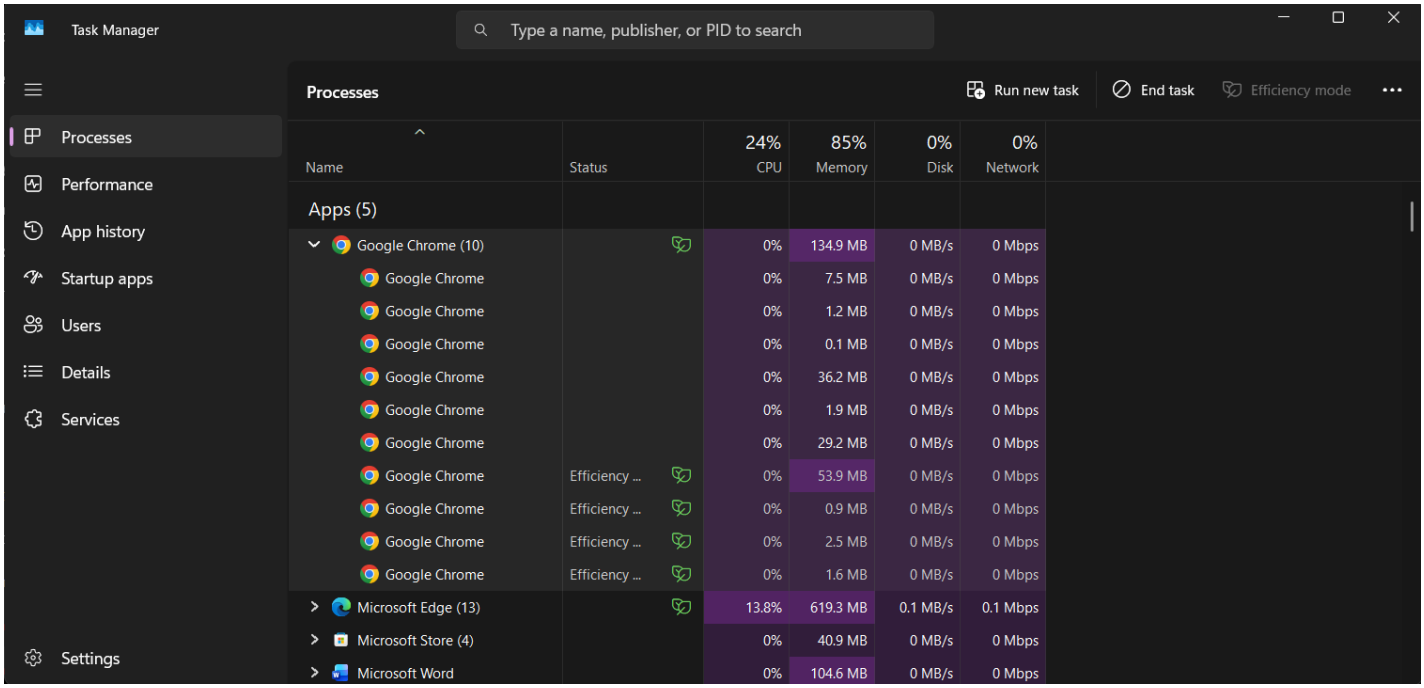
➢ **Introduction**

- **Task Manager** is a built-in Windows utility used to monitor running applications, background processes, CPU usage, memory usage, and other system resources.
- It helps users understand how much RAM is being used by different applications and processes.
- In this practical, Task Manager is used to identify processes related to Google Chrome, WhatsApp Desktop, or other browser-based applications and record their memory consumption.

➢ **Requirements**

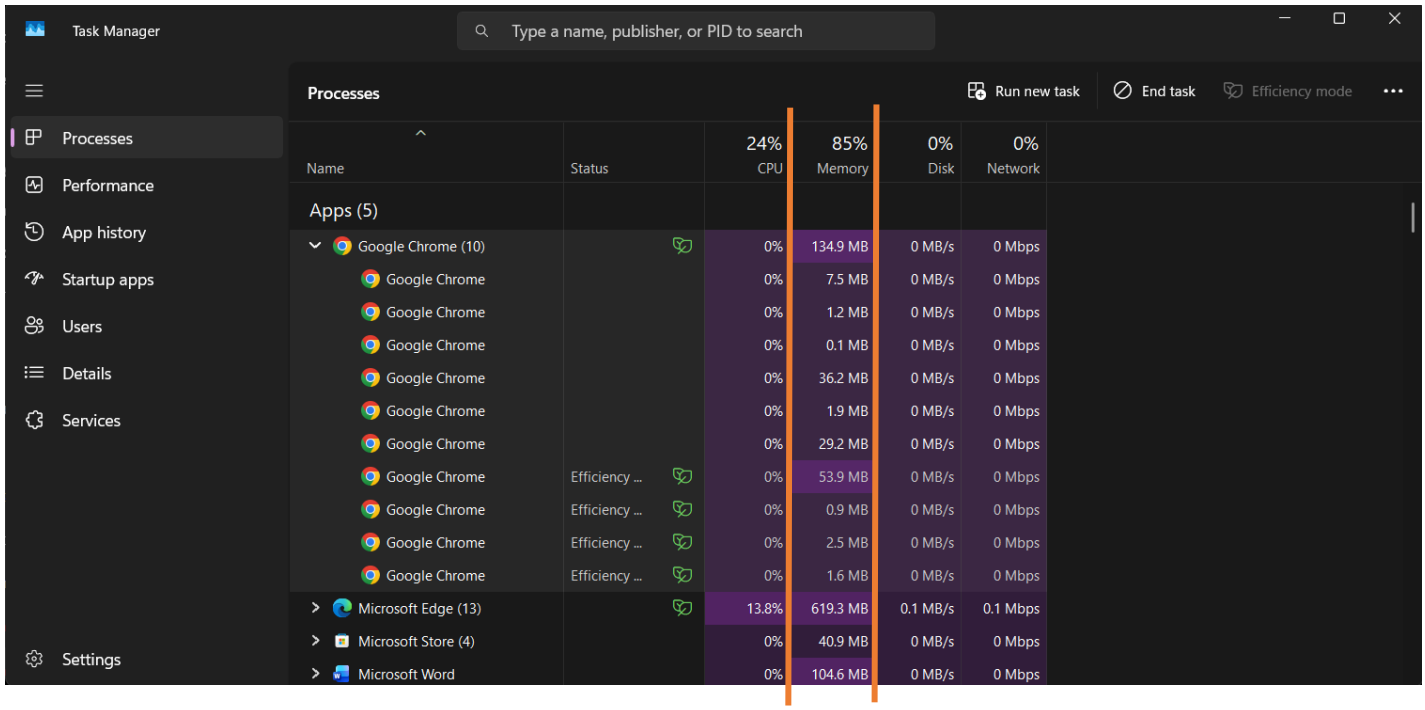
- A Windows computer or laptop
- Windows operating system
- Google Chrome, WhatsApp Desktop, or another browser-based application
- Task Manager

- **Procedure**
- **Step 1: Open Task Manager**
- Press Ctrl + Shift + Esc on the keyboard.
- The Windows Task Manager window will open.
- Click **More details** if the simplified view appears.
- **Step 2: Open the Processes Tab**
- Select the **Processes** tab.
- Observe the list of running applications and background processes.
- Look for Google Chrome, WhatsApp Desktop, or other browser-based applications.



- **Step 3: Check Memory Usage**
- Locate the **Memory** column.
- Observe the memory usage displayed for each relevant process.

- If the Memory column is not visible, use the Details tab or adjust the Task Manager columns.
- Record the process names and their memory usage.



- **Step 4: Record the Observations**
- Prepare a table containing the process name and its corresponding memory usage.

- **Observation**
- The following table should be completed using the actual information displayed in Windows Task Manager.

Sr. No.	Process Name	Memory Usage
1	Google Chrome (chrome.exe)	134.9 MB
2	Microsoft Edge (msedge.exe)	619.3 MB

- **Result**
- Windows Task Manager was used to identify running processes related to Google Chrome, WhatsApp Desktop, or other browser-based applications. Their memory usage was observed and recorded.

❖ **4. Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network. Make sure your firewall allows scanning, and only scan your own home/college network.**

- **Aim**
  - To use Advanced IP Scanner to scan a local Wi-Fi network and identify the IP addresses and device names of at least two connected devices.

- **Introduction**
  - **Advanced IP Scanner** is a free network scanning tool used to discover devices connected to a local network.

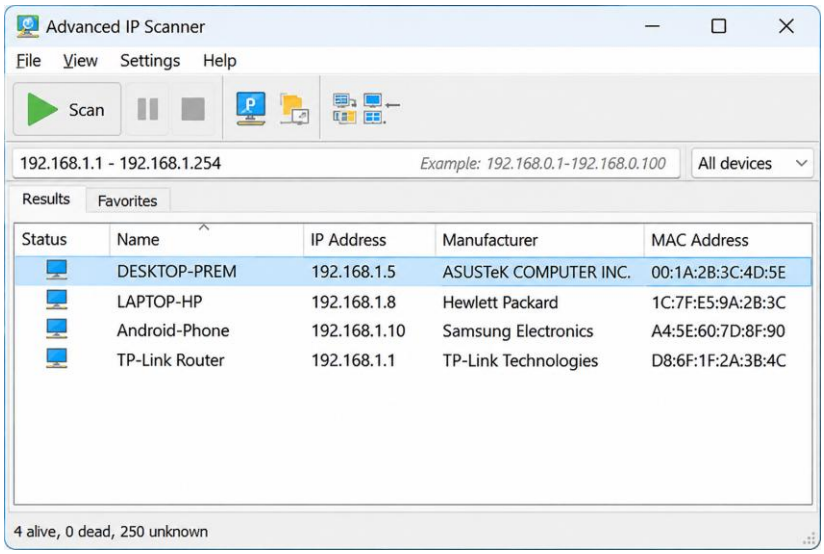
- It scans a specified IP address range and displays information about detected devices, such as their IP addresses, device names, and MAC addresses.
- In this practical, Advanced IP Scanner is used to scan our own Wi-Fi network and identify at least two connected devices.

➢ **Requirements**

- A Windows laptop or computer
- Active Wi-Fi connection
- Advanced IP Scanner software
- Permission to scan the local network

➢ **Procedure**

1) Download Advanced IP Scanner from its official website.
2) Install and open the software.
3) Connect the laptop to your own home or college Wi-Fi network.
4) Allow the software to detect the local IP address range.
5) Check that the selected range belongs to your own local network.
6) Click the **Scan** button to start scanning.
7) Wait until the scan is completed.
8) Examine the list of discovered devices.
9) Identify at least two devices and note their IP addresses and device names.



10) Save or capture a screenshot of the scan results.

- **Observation**
- The following table should be completed using the actual results displayed by Advanced IP Scanner.

Sr. No.	Device Name	IP Address
1	DESKTOP-PREM	192.168.1.5
2	LAPTOP-HP	192.168.1.8

- **Result**
- Advanced IP Scanner was used to scan the local Wi-Fi network. At least two connected devices were identified, and their IP addresses and device names were recorded.

❖ **5. Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.**

➢ **Aim**

- To identify three hardware tools used in network troubleshooting and explain their uses.

➢ **Introduction**

- Hardware tools play an important role in identifying and solving network-related problems. They help network technicians test cables, check electrical connections, and detect faults in networking equipment.
- The following are three commonly used hardware tools in network troubleshooting.

➢ **Three Hardware Tools and Their Uses**

Sr. No.	Hardware Tool	When It Is Used
1	Cable Tester	A cable tester is used to check Ethernet cables for broken wires, incorrect wiring, and connection faults.

2	Crimping Tool	A crimping tool is used to attach RJ-45 connectors to Ethernet cables when preparing or repairing network cables.
3	Multimeter	A multimeter is used to measure voltage, resistance, and continuity when checking electrical faults in networking equipment or power connections.

- **Explanation**
- **1. Cable Tester**

  A cable tester checks whether all the wires inside an Ethernet cable are connected correctly. It helps identify problems such as open circuits, short circuits, and incorrect wire connections.
- **2. Crimping Tool**

  A crimping tool is used to attach an RJ-45 connector to the end of an Ethernet cable. It is useful when making new network cables or replacing damaged connectors.
- **3. Multimeter**

  A multimeter is an electronic measuring instrument used to check voltage, resistance, and electrical continuity. It helps technicians diagnose electrical problems in networking devices and their power supplies.

- **Result**
- Three hardware tools used in network troubleshooting—Cable Tester, Crimping Tool, and Multimeter—were identified, and their uses were studied.